

Grade 6 Accelerated
Unit 11
Lesson 8 Homework

Name Answer Key
Date _____
Period _____

1. Emily says that when she evaluated an expression, the result was 51. Use mental math to estimate each value. Then, circle which of the following expressions could have been Emily's expression.

$$(8 + 3)^2 - 22$$

$$8^2 + 3^2 - 22$$

$$(8 - (-3)^2) + 22$$

$$11^2 + 22$$

2. Cindy evaluated the expression $5^2 \div 5(3.25 + 5.25)$.

Circle Cindy's error and choose which step of the order of operations she did not apply correctly.

Cindy's Steps	Order of Operations
$5^2 \div 5(3.25 + 5.25)$ $10 \div 5(8.5)$ $2(8.5)$ 17	• Evaluate expressions with exponents • Perform operations inside grouping symbols. • Perform multiplication and division in order from left to right. • Perform addition and subtraction in order from left to right.

3. Evaluate the expression without using a calculator. Write an explanation using the order of operations next to each step.

$ 4 - 12 \div (-8)$	Explanation
$ -8 \div (-8)$	Grouping symbol
$8 \div (-8)$	Division
-1	

4. Evaluate the expression without using a calculator. Write an explanation using the order of operations next to each step.

$(-3.5 \times 4)^2 \div 7$	Explanation
$(-14)^2 \div 7$	Grouping symbol
$196 \div 7$	Exponents
28	Division

Evaluate each expression.

5. $\frac{(-14 \times 0.5)^3}{14} = \frac{-7^3}{14} = \frac{-343}{14} = -24.5$

6. $6 \times |-3.4 \times 5.2| \quad 6 \times |-17.68| = 6 \times 17.68 = 106.08$

7. $(-84 \times 0.6)^2 + 6 \quad (-50.4)^2 + 6 = 2540.16 + 6 = 2546.16$

8. $[(16 - 5 \times 4) \times 3]^3$

$$[(16 - 20) \times 3]^3$$

$$[-4 \times 3]^3$$

$$-12^3$$

$$-1,728$$

9. $64 \div [(-3)^2 \times 4 - 2]$

$$64 \div [9 \times 4 - 2]$$

$$64 \div [36 - 2]$$

$$64 \div 34$$

$$1 \frac{15}{17}$$

10. $|-6.3 \times 5| - 5^2$

$$|-31.5| - 25$$

$$31.5 - 25$$

$$6.5$$